
CF08: COMPLETE FORMALISM — SPINOR EMERGENCE VIA DOMAIN-WALL FERMIONS IN VDM (NIELSEN–NINOMIYA DEFENSE)

COMPLETE FORMALISM

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Abstract

Lattice discretizations of chiral fermions face a classical obstruction: under mild assumptions (locality, translational invariance, and a single lattice Dirac operator), the Nielsen–Ninomiya theorem forces species doubling. This CF adapts a standard evasion mechanism—domain-wall fermions and the overlap/Ginsparg–Wilson construction—to the Void Dynamics Model (VDM) matter-sector program. An auxiliary lattice coordinate z supports a sign-changing mass profile (a kink/domain wall) whose exponentially localized zero modes behave as low-momentum chiral spinors on the $3 + 1$ D wall. Integrating out the gapped bulk yields an effective overlap operator satisfying the Ginsparg–Wilson relation, providing an exact lattice-scale notion of chiral symmetry and a quantitative residual for finite wall separation. Two additional attack surfaces are addressed as explicit gates: (i) residual mass (chirality leakage) must decay exponentially with wall separation, and (ii) any fermionization/encoding used for simulation must preserve locality at most polylogarithmically in system size (Bravyi–Kitaev rather than $\mathcal{O}(N)$ Jordan–Wigner strings). The deliverable is a decisive, auditable gate suite (P1–P5) and an experiment plan for the companion notebook CFN implementation.

Keywords lattice fermions · domain-wall fermions · overlap operator · Ginsparg–Wilson relation · Nielsen–Ninomiya theorem · Bravyi–Kitaev transformation · chirality · locality · Void Dynamics Model

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1 Objective, scope, and non-claims

The Void Dynamics Model (VDM) matter-sector program (Gap Module S5) requires a mechanism by which *spinorial* low-energy degrees of freedom can be represented on top of a microscopic lattice description whose conservative J-limb is specified in terms of scalar lattice fields [Lietz, 2025a,b]. The immediate obstacle is not numerical but structural: a naive lattice Dirac operator for chiral fermions typically suffers *species doubling*.

Claim: S0 (Problem statement: Nielsen–Ninomiya obstruction) *Decisive metric: absence of additional chiral zero modes at Brillouin-zone corners.*

Under standard assumptions (locality, translational invariance, and a single lattice Dirac operator), the Nielsen–Ninomiya theorem implies that chiral fermions on a lattice come with doublers [Nielsen and Ninomiya, 1981]. Any VDM matter-sector construction that purports to yield chiral fermions must explicitly identify which assumption is relaxed and must show (by an auditable metric) that extra low-energy modes are absent.

Approach. This CF uses a standard, well-tested evasion mechanism: *domain-wall fermions* [Kaplan, 1992, Shamir, 1993] and the induced *overlap/Ginsparg–Wilson* operator [Ginsparg and Wilson, 1982, Neuberger, 1998]. The core idea is to introduce an auxiliary lattice coordinate z supporting a sign-changing mass profile (a kink/domain wall). Chiral zero modes localize on the wall and are separated topologically from would-be doublers in the gapped bulk.

Deliverable. The deliverable is not a “physics claim” by itself; it is a publishable *formal construction* plus a decisive gate suite (P1–P5) that a companion notebook CFN must implement.

Scope classification. The module is classified as *derived-limit*: it imports standard lattice-fermion constructions as an intermediate effective description, then binds them to VDM-specific observables and gates [Lietz, 2025c,b].

Non-claims (tight).

- This CF does *not* claim novelty for domain-wall, overlap, or Ginsparg–Wilson mathematics; these are classical lattice-QCD tools [Kaplan, 1992, Shamir, 1993, Ginsparg and Wilson, 1982, Neuberger, 1998].
- This CF does *not* by itself prove that fermionic *exchange statistics* emerge from a purely scalar microscopic state. If an explicit “minus sign under exchange” mechanism is required in VDM, it is a separate topological/statistics module with its own decisive gates.
- This CF does *not* assume Lorentz invariance at the lattice scale; instead it specifies a measurable low-momentum isotropy gate.

Deeper foundation in CF13. When CF08 was first written, the Nielsen–Ninomiya obstruction was treated as a classical lattice result to be evaded by construction. CF13 [Lietz, 2026a] now provides the invariant-geometric explanation: chirality is the 5D effective invariant of oriented boundary irreconcilability, and the obstruction is the statement that the transcendental constant π cannot be contained within finite periodic algebraic closure. The domain-wall construction of this paper is the correct response to that obstruction: orthogonal re-articulation into an auxiliary dimension that asymptotically approaches the transcendental boundary rather than attempting to resolve it algebraically. See CF13 §7–8.

2 Claim inventory and decisive falsifiers

This paper treats claims as contracts: each claim has a decisive metric and a falsifier.

2.1 Primary claims

Claim: C1 (Domain-wall chiral zero mode) *Decisive metric: existence of a normalizable exponentially localized mode with definite chirality.*

A sign-changing mass profile in an auxiliary coordinate z supports exponentially localized low-energy modes bound to the domain wall. These modes are the *only* degrees of freedom kept as “physical” in the effective wall theory; bulk modes must remain gapped.

Claim: C2 (Linear Dirac dispersion) *Decisive metric: R^2 of linear fit $E(p) = v_F |p|$ exceeds threshold on a declared low-momentum window.*

In the low-momentum regime $|p| \ll \pi/a$, the bound wall mode exhibits linear dispersion with controlled lattice corrections.

Claim: C3 (Overlap/Ginsparg–Wilson effective operator) *Decisive metric: $\|\{D, \gamma_5\} - aD\gamma_5 D\|$ below tolerance.*

Integrating out the gapped bulk yields an effective wall Dirac operator D satisfying the Ginsparg–Wilson relation (exactly in the infinite wall-separation limit; approximately with a quantified residual at finite wall extent).

Claim: C4 (Local encoding for simulation) *Decisive metric: fermion operator support scales at most polylogarithmically in system size.*

Any encoding used to realize fermionic operators on a qubit/spin (or scalar-site) substrate must preserve locality up to polylogarithmic overhead (Bravyi–Kitaev rather than $\mathcal{O}(N)$ Jordan–Wigner strings).

2.2 Assumption ledger

The following assumptions are explicit and falsifiable:

1. **A1 (Effective Dirac factorization):** the relevant linearized operator extracted from VDM coarse degrees of freedom can be represented as a first-order Dirac-type operator on an internal (Clifford) space.
2. **A2 (Stable bulk gap):** coarse-graining produces a stable sign-changing “mass” profile and a gapped bulk spectrum.
3. **A3 (Direct testing in CFN):** the companion notebook tests the chiral mode, dispersion, and GW residual directly; if any gate fails, the emergence claim is false for that model instance.

2.3 Decisive gates (P1–P5)

The decisive falsifiers are implemented as gates (thresholded metrics) adopted from the hypothesis H005 [Lietz, 2025b]:

Gate: P1 (Ginsparg–Wilson residual) *Threshold: $\|\{D, \gamma_5\} - aD\gamma_5 D\|_\infty \leq 10^{-12}$ on coarse cells $\ell = 4a$.*

If the GW residual exceeds tolerance, the effective operator is not chiral-protected at the lattice scale and C3 fails.

Gate: P2 (Residual mass suppression) *Threshold: $m_{\text{res}}(L_5)/m_{\text{res}}(L_5/2) \leq e^{-\lambda L_5/2}$ with $\lambda \geq 0.1/a$.*

If the residual mass fails to decay exponentially with wall separation, chirality protection is not practically attainable and the domain-wall mechanism is not viable.

Gate: P3 (Dispersion linearity) *Threshold: for $|p| < \pi/(10a)$, linear fit $R^2 \geq 0.9999$.*

If the dispersion is not linear on the declared low-momentum window, the emergent mode is not Dirac-like and C2 fails.

Gate: P4 (Rotational isotropy) *Threshold: angular variation $\Delta E/\bar{E} \leq 10^{-3}$ at fixed $|p| = 0.1\pi/a$.*

If isotropy fails, the low-energy sector violates the intended Lorentz/rotational symmetry to a degree that must be treated as a falsifier for Standard-Model-style use.

Gate: P5 (Encoding locality) *Threshold: operator support $\leq C \log^2 N$ over $N \in \{32^3, 64^3, 128^3\}$.*

If locality scaling is worse than polylogarithmic, the encoding violates the VDM locality/causality discipline and C4 fails.

3 Foundations: VDM scalar lattice and domain-wall setting

3.1 J-limb scalar lattice action and continuum limit

The conservative VDM J-limb admits a scalar lattice action (with discrete time index n and lattice site index i) [Lietz, 2025a, 2026b,c]:

$$S_{\text{lattice}} = \sum_{n,i} \left[\frac{1}{2} \left(\frac{W_i^{n+1} - W_i^n}{\Delta t} \right)^2 - \frac{J}{2} \sum_{j \in N(i)} (W_j^n - W_i^n)^2 - V(W_i^n) \right] \Delta t. \quad (1)$$

In a standard long-wavelength limit (lattice spacing a , field $\phi(x, t)$), the continuum action takes the Klein–Gordon-like form

$$S_{\text{cont}} = \int dt d^3x \left[\frac{1}{2} (\partial_t \phi)^2 - \frac{c^2}{2} |\nabla \phi|^2 - V(\phi) \right], \quad c^2 = 2Ja^2. \quad (2)$$

A prototypical “tachyonic” potential used to admit kink solutions is

$$V(\phi) = -\frac{1}{2} \mu^2 \phi^2 + \frac{1}{4} \lambda \phi^4, \quad \mu^2 > 0, \lambda > 0, \quad (3)$$

with vacua $\phi_{\pm} = \pm \mu / \sqrt{\lambda}$.

3.2 Auxiliary coordinate and domain-wall interpretation

The domain-wall construction introduces an auxiliary coordinate z (continuous or discretized as $z_k = ka_5$ for $k \in \{0, \dots, L_5\}$). The physical effective degrees of freedom are identified with modes localized near a chosen wall, taken here to be at $z = 0$. The auxiliary dimension is a *construction device*: bulk excitations are required to remain gapped so that the wall theory is effectively 3 + 1D. The 5D auxiliary coordinate is the orthogonal re-articulation axis forced when the 4D chiral degree of freedom saturates; see CF00 [Lietz, 2026c] for the derivation of the carrier and differentiable structure that makes this construction lawful, and CF000 [Lietz, 2026b] for the primitive root from which orthogonal re-articulation is forced.

Remark 1 (How the construction evades Nielsen–Ninomiya). The Nielsen–Ninomiya theorem constrains chirality on a single translationally invariant lattice Dirac operator [Nielsen and Ninomiya, 1981]. Domain-wall fermions evade the obstruction by enlarging the system (adding z) and breaking translational invariance along z via a topological defect; physical modes are confined to the defect rather than occupying the full bulk [Kaplan, 1992, Shamir, 1993].

Remark 2 (Nielsen–Ninomiya as transcendence obstruction — CF13). CF13 supplies a deeper explanation of why the domain-wall construction takes the form it does. Exact chirality lives at the half-turn angle $\pi := \inf\{\theta > 0 : R_\theta = -I\}$, the minimum rotation required to reach the opposite pole of the primitive bifurcation invariant [Lietz, 2026a]. Because π is transcendental over $\mathbb{Q}$ — derived from the non-discharge condition of CF000 alone, with Lindemann’s theorem recovered as a corollary — the finite periodic algebraic closure of a 4D lattice cannot contain π exactly. Nielsen–Ninomiya is therefore not a random lattice accident; it is the algebraic-closure obstruction to isolating one oriented half-turn sector inside finite periodic closure. The domain-wall construction evades this obstruction not by violating the theorem’s hypotheses but by asymptotically separating the unwanted chiral partner in the auxiliary dimension z , where the exact transcendental boundary is approached rather than finitely resolved. This paper is the constructive paper; CF13 is the explanatory paper.

3.3 Symbol inventory (minimal)

Symbol	Meaning	Notes
W_i^n	scalar lattice field	VDM J-limb variable
$\phi(x, t)$	continuum scalar field	long-wavelength limit
a	spatial lattice spacing	fundamental length scale
a_5	auxiliary (bulk) lattice spacing	in z direction
L_5	bulk depth (in units of a_5)	wall separation control
ψ	emergent wall spinor field	effective, low-energy
γ^μ, γ_5	Clifford matrices	standard conventions
D	effective lattice Dirac operator	wall theory

Symbol	Meaning	Notes
m_{res}	residual mass	chirality leakage metric
λ	wall decay rate	controls exponential suppression
v_F	Fermi velocity	slope of dispersion

4 Domain-wall background and kink profile

4.1 Continuous kink profile and exponential tails

A static kink profile $\phi_{\text{bg}}(z)$ is a solution of the one-dimensional Euler–Lagrange equation

$$-c^2 \frac{d^2 \phi_{\text{bg}}}{dz^2} + V'(\phi_{\text{bg}}) = 0. \quad (4)$$

For the quartic potential Equation (3), the standard kink solution is

$$\phi_{\text{bg}}(z) = \phi_+ \tanh\left(\frac{z - z_0}{\xi}\right), \quad \xi = \frac{c}{\mu}, \quad \phi_+ = \frac{\mu}{\sqrt{\lambda}}, \quad (5)$$

with exponential approach to the vacua

$$|\phi_{\text{bg}}(z) - \phi_+| \sim 2\phi_+ e^{-|z|/\xi} \quad (|z| \gg \xi), \quad (6)$$

so the decay rate is

$$\lambda \equiv \xi^{-1} = \mu/c. \quad (7)$$

4.2 Discrete kink on a finite z -lattice

On a discretized auxiliary dimension $z_k = ka_5$, the second derivative in Equation (4) is replaced by a finite difference:

$$-c^2 \frac{\phi_{k+1} - 2\phi_k + \phi_{k-1}}{a_5^2} + V'(\phi_k) = 0. \quad (8)$$

A practical construction uses boundary conditions $\phi_0 \approx -\phi_+$, $\phi_{L_5} \approx \phi_+$ and a relaxation/Newton method to obtain $\{\phi_k\}$.

Gate: G0 (Domain-wall fit preflight) *Threshold: exponential tail fit $R^2 \geq 0.99$ and $\lambda \geq 0.1/a$.*

This gate is a preflight for P2: if the wall mode is not localized (small λ) or the profile is not kink-like, residual-mass suppression cannot be expected.

5 Bound-state spectrum and wall-localized chiral modes

5.1 Linearization and a Schrödinger-type problem in z

Write $\phi(x, z, t) = \phi_{\text{bg}}(z) + \delta\phi(x, z, t)$. To quadratic order, fluctuations obey

$$S_{\text{lin}} = \int dt d^3x dz \left[\frac{1}{2}(\partial_t \delta\phi)^2 - \frac{c^2}{2}(|\nabla \delta\phi|^2 + |\partial_z \delta\phi|^2) - \frac{1}{2}V''(\phi_{\text{bg}}) \delta\phi^2 \right]. \quad (9)$$

Separating variables $\delta\phi(x, z, t) = \chi(z) e^{i(\mathbf{p} \cdot \mathbf{x} - Et)}$ yields a bound-state equation in z :

$$\left[-c^2 \frac{d^2}{dz^2} + V''(\phi_{\text{bg}}(z)) \right] \chi(z) = \epsilon \chi(z), \quad (10)$$

with wall-parallel dispersion

$$E^2(\mathbf{p}) = \epsilon + c^2 |\mathbf{p}|^2. \quad (11)$$

5.2 Zero mode and linear dispersion

For the kink background, the standard translational zero mode is proportional to the derivative of the kink:

$$\chi_0(z) \propto \frac{d\phi_{\text{bg}}}{dz} \sim \text{sech}^2(z/\xi), \quad (12)$$

and is normalizable. When the effective low-energy sector factors into a Dirac-type operator on an internal space (Assumption A1), this wall-localized mode corresponds to a chiral spinor degree of freedom.

Theorem 1 (Low-momentum linear dispersion for the wall mode). *Assume $\epsilon_0 \approx 0$ for the lowest wall mode in Equation (11). Then for $|\mathbf{p}| \ll \pi/a$,*

$$E(\mathbf{p}) = v_F |\mathbf{p}| + \mathcal{O}(|\mathbf{p}|^3), \quad v_F = c + \mathcal{O}(a^2). \quad (13)$$

Gate: P3 (Dispersion linearity) *Threshold: $R^2 \geq 0.9999$ for $|p| < \pi/(10a)$.*

The CFN implementation must fit $E(p)$ to Equation (13) on the declared window and report the fit residuals.

6 Overlap/Ginsparg–Wilson operator and chiral symmetry protection

6.1 From domain walls to overlap

A naive Wilson discretization introduces a “Wilson term” that removes doublers but explicitly breaks chiral symmetry. The lattice-QCD resolution is to require a modified chiral symmetry encoded by the Ginsparg–Wilson relation [Ginsparg and Wilson, 1982]:

$$\{D, \gamma_5\} = a D \gamma_5 D. \quad (14)$$

In the standard construction, domain-wall fermions in $4 + 1$ dimensions induce an effective $3 + 1\text{D}$ *overlap* operator in the infinite-wall limit [Kaplan, 1992, Shamir, 1993, Neuberger, 1998]. Let D_W be a Wilson–Dirac operator on the wall lattice directions and define $H_W = \gamma_5 D_W$. The overlap operator can be written

$$D_{\text{ov}} = \frac{1}{a} \left(1 + \gamma_5 \text{sign}(H_W) \right). \quad (15)$$

A common equivalent form (used in some conventions) is

$$D_{\text{ov}} = \frac{1}{a} \left(1 - \frac{H_W}{\sqrt{H_W^\dagger H_W}} \right), \quad (16)$$

which makes explicit that the construction uses the polar decomposition of H_W .

Theorem 2 (Ginsparg–Wilson identity for the overlap operator). *The overlap operator Equation (15) satisfies the Ginsparg–Wilson relation Equation (14) exactly.*

Proof sketch. Use $\gamma_5^2 = 1$ and $\text{sign}(H_W)^2 = 1$. Substitute Equation (15) into both sides of Equation (14) and simplify algebraically; all terms reduce to identities involving $\text{sign}(H_W)$. $\square$

6.2 Finite wall separation and the GW residual

At finite wall extent L_5 , the sign function is approximated by a transfer-matrix/rational approximation, producing an exponentially small defect controlled by the bulk gap λ . Discretization also induces $\mathcal{O}(a^2)$ long-wavelength error. A convenient auditable residual bound is

$$\|\{D, \gamma_5\} - a D \gamma_5 D\|_\infty \leq C_1 a^2 + C_2 e^{-\lambda L_5}, \quad (17)$$

with constants C_1, C_2 set by the chosen coarse-graining and bulk gap.

Gate: P1 (Ginsparg–Wilson residual) *Threshold: $\|\{D, \gamma_5\} - a D \gamma_5 D\|_\infty \leq 10^{-12}$.*

The CFN implementation must construct D (or D_{ov} directly) and report the operator-norm residual in Equation (14).

6.3 Residual mass and chirality leakage

For finite wall separation, left- and right-handed modes localized on separated walls can mix through tunneling in the gapped bulk, inducing a small *residual mass* m_{res} . A simple overlap estimate yields exponential suppression.

Theorem 3 (Exponential suppression of residual mass). *Let $\chi_L(z)$ and $\chi_R(z)$ be normalized wall wavefunctions separated by L_5 with exponential tails bounded by $e^{-\lambda z}$. Then the leading mixing amplitude obeys*

$$\langle \chi_L, \chi_R \rangle \lesssim L_5 e^{-\lambda L_5}, \quad (18)$$

and therefore

$$m_{\text{res}} \sim A e^{-\lambda L_5} \quad (19)$$

for a prefactor A set by the bulk mass scale.

Proof sketch. Bound $|\chi_L(z)\chi_R(z)| \lesssim e^{-\lambda z} e^{-\lambda(L_5-z)}$ and integrate over $[0, L_5]$ to obtain Equation (18). The induced chirality-flipping term in the effective action is proportional to this overlap. $\square$

Gate: P2 (Residual mass suppression) *Threshold:* $m_{\text{res}}(L_5)/m_{\text{res}}(L_5/2) \leq e^{-\lambda L_5/2}$.

The CFN implementation must measure m_{res} over a declared set of L_5 values and verify exponential scaling consistent with Equation (19).

7 Locality and Bravyi–Kitaev fermionization

7.1 The Jordan–Wigner locality problem in higher dimensions

A standard Jordan–Wigner mapping in 1D expresses fermion operators as spin operators multiplied by a parity string. In 3D, any naive total ordering induces parity strings that can span $\mathcal{O}(N)$ sites, making operator support nonlocal and conflicting with locality/causality constraints.

7.2 Bravyi–Kitaev transformation

The Bravyi–Kitaev transformation replaces the long parity string with a binary-tree parity bookkeeping structure [Bravyi and Kitaev, 2002]. For N modes, operator support is reduced from $\mathcal{O}(N)$ to polylogarithmic.

Definition 1 (Bravyi–Kitaev encoding (operational description)). Label sites by an index $i \in \{0, \dots, N-1\}$ (e.g. via a space-filling curve). Construct a binary tree over indices. Define parity and update sets $P(i), U(i)$ from the tree ancestry. Then each fermion operator can be represented using products over these sets, yielding $|P(i)|, |U(i)| = \mathcal{O}(\log N)$ and overall support $\mathcal{O}(\log^2 N)$ in 3D embeddings.

Gate: P5 (Encoding locality) *Threshold:* $\text{support} \leq C \log^2 N$ over the declared sizes.

The CFN implementation must report support statistics (max/median) for representative creation/annihilation operators and confirm the scaling law.

7.3 Causality check: mode velocity must not scale with system size

Even if operator *support* is polylogarithmic, information propagation in the effective theory must remain consistent with an N -independent velocity scale. The dispersion slope v_F from Equation (13) provides an auditable proxy: v_F must not drift systematically with lattice size when the physical momentum window is held fixed.

8 Low-energy isotropy and Lorentz-compatibility gate

8.1 Hypercubic anisotropy at the lattice scale

On a cubic lattice, high-momentum dispersion relations inherit only hypercubic symmetry rather than full rotational symmetry. For example, simple lattice Laplacians yield behaviors such as

$$E^2(\mathbf{p}) \propto \sum_{\mu=1}^3 \sin^2(p_\mu a), \quad (20)$$

which is anisotropic near the Brillouin-zone boundary.

8.2 Operational isotropy requirement

This CF does not require exact Lorentz invariance at the lattice scale. Instead, it requires that the *low-energy* wall mode exhibits small angular variation at fixed $|\mathbf{p}|$ in a declared window. This is formalized by P4.

Gate: P4 (Rotational isotropy) *Threshold:* $\Delta E/\bar{E} \leq 10^{-3}$ at $|p| = 0.1\pi/a$.

Compute $E(\mathbf{p})$ on a set of angles (e.g. 24 directions) at fixed $|\mathbf{p}|$. Define $\bar{E}$ as the mean over angles and ΔE as max-minus-min. The relative variation must be below threshold.

8.3 Role of metriplectic coarse-graining (interpretive note)

Within VDM, the dissipative M-limb is interpreted as a coarse-grained “shadow” of the conservative J-limb, with degeneracy conditions enforcing compatibility between conserved and dissipative generators [Lietz, 2025a,d]. In this interpretation, M-limb coarse-graining can act as a low-pass filter that suppresses lattice-scale anisotropy in effective observables. This note is interpretive and does not replace the operational P4 gate.

9 Summary: defenses against the three red-team attack vectors

The motivating red-team objections and the corresponding defenses in this CF are:

1. **Non-local Jordan–Wigner strings:** mitigated by Bravyi–Kitaev encoding with polylogarithmic support (P5) [Bravyi and Kitaev, 2002].
2. **Chiral symmetry leakage via residual mass:** quantified by exponential residual-mass suppression with wall separation (P2).
3. **Lorentz violation / lattice anisotropy:** treated as a low-energy isotropy requirement (P4) rather than a lattice-scale assumption.

The key structural element is the overlap/Ginsparg–Wilson operator: chiral symmetry is preserved in a lattice-consistent way (P1) [Ginsparg and Wilson, 1982, Neuberger, 1998].

10 Validation gate summary table

Gate	Observable	Comparator	Threshold
P1	$\ \{D, \gamma_5\} - aD\gamma_5 D\ _\infty$	$\leq$	10^{-12}
P2	$m_{\text{res}}(L_5)/m_{\text{res}}(L_5/2)$	$\leq$	$e^{-\lambda L_5/2}$
P3	R^2 for $E(p) = v_F p $ fit	$\geq$	0.9999
P4	$\Delta E/\bar{E}$ at fixed $ p $	$\leq$	10^{-3}
P5	BK operator support	$\leq$	$C \log^2 N$

Table 2: Decisive gate suite adopted from H005 (spinor emergence Nielsen–Ninomiya defense) [Lietz, 2025b].

11 Companion notebook outputs and experiment plan

The companion notebook CFN 8 (pending in the repository) is required to implement a 1:1 recreation of this CF and produce auditable artifacts. The minimal experiment plan is:

1. **E1 (Domain-wall profile):** construct $\phi_{\text{bg}}(z)$, fit λ , pass preflight G0.
2. **E2 (Bound-state spectrum):** compute the lowest wall mode and its dispersion; enforce P3.
3. **E3 (GW verification):** construct D (or D_{ov}) and enforce P1.
4. **E4 (Residual mass scaling):** measure m_{res} versus L_5 ; enforce P2.

5. **E5 (Isotropy)**: evaluate angular variation at fixed $|p|$; enforce P4.
6. **E6 (BK locality scaling)**: measure operator support vs system size; enforce P5.

All runs must log deterministic seeds, configuration manifests, and gate results under the VDM validation-metrics discipline [Lietz, 2025c].

12 References discipline

External references are limited to the standard lattice-fermion and fermionization literature required to support the mathematical construction [Nielsen and Ninomiya, 1981, Ginsparg and Wilson, 1982, Kaplan, 1992, Shamir, 1993, Neuberger, 1998, Bravyi and Kitaev, 2002]. VDM-internal documents (axioms, equation registry, validation metrics, and the H005 hypothesis) are cited as repository snapshots [Lietz, 2025a,d,e,c,b]. CF13 [Lietz, 2026a] is additionally cited as the explanatory paper that derives the Nielsen–Ninomiya obstruction as a transcendence obstruction from the primitive bifurcation invariant, and derives chirality as the 5D effective invariant of oriented boundary irreconcilability.

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